

Attention Mechanisms in Neural Networks: A Survey

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Abstract

We survey attention mechanisms in deep learning, covering transformers, self-attention, and cross-attention architectures. We analyze 200 papers.

Keywords: attention, transformers, neural networks, survey

1 Introduction

Attention mechanisms have become a cornerstone of modern deep learning. First introduced by Bahdanau et al. (2015) for machine translation, they have since been adopted across vision, language, and multi-modal tasks. The transformer architecture (Vaswani et al., 2017) relies exclusively on attention, dispensing with recurrence entirely.

2 Background

Given query Q , keys K , and values V , the scaled dot-product attention is $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$, where d_k is the key dim. Multi-head attention projects Q, K, V into h parallel heads, concatenates the outputs, and projects back to model dimension.

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3 Related Work

Luong et al. (2015) proposed multiplicative attention as a cheaper alternative. Transformer-XL (Dai et al., 2019) extends the context window via segment-level recurrence. BERT (Devlin et al., 2019) uses bidirectional self-attention for pre-training on masked language modelling.

Acknowledgements

The authors thank the MIT CSAIL computing cluster for GPU resources. This work was supported by NSF grant IIS-2345678 and a Google Faculty Award. Carol Wu was supported by an MIT Presidential Fellowship.

References

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